GHOST: Generalizing Skills By Hierarchical Learning from Human Demonstrations

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https://ghost-human-demo.github.io/

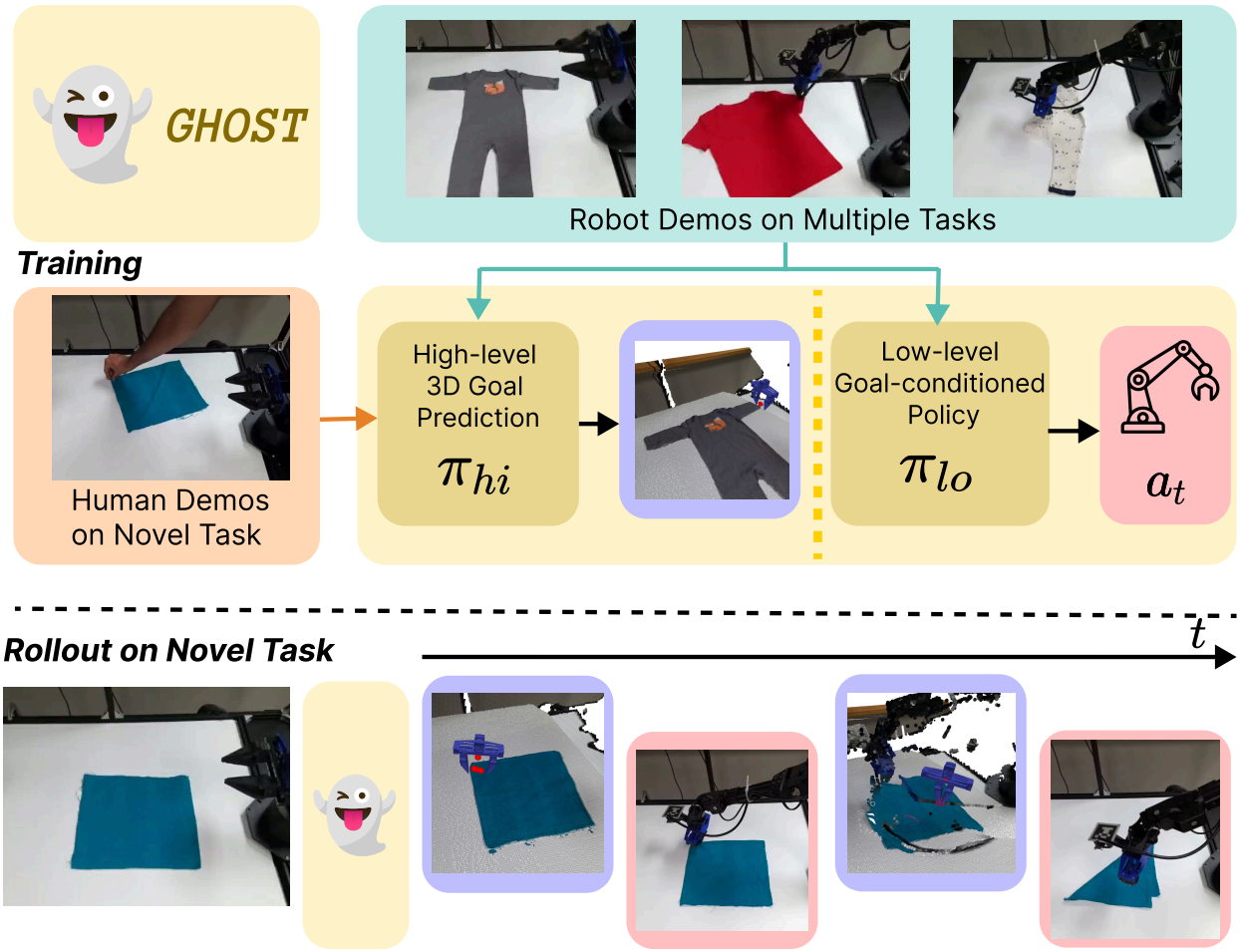

Fig. 1: GHOST learns robot skills from in-distribution teleoperation data and generalizes to novel tasks using human demonstrations. We train a hierarchical policy that decouples task-specific goal prediction (π_{hi}) from embodiment-specific action execution (π_{lo}). By training π_{hi} on both robot and human data while keeping π_{lo} purely on robot data, GHOST transfers learned manipulation skills to out-of-distribution tasks with only human video demonstrations.

***Abstract*—We present GHOST, a framework for generalizing robot skills to new tasks through human demonstrations. Robot teleoperation data is critical for training robust policies, but is expensive to scale, while human video is abundant, but lacks executable robot actions. We use a hierarchical policy framework that decouples a policy into two parts. The high-level policy maps visual observations to task "sub-goals". These sub-goals are intermediate objectives like "grasp the handle" defined in a task-relevant but embodiment-agnostic manner, and do not require action labels. The low-level policy is conditioned on these goals to execute embodiment-specific actions and is trained purely on robot data. This setup provides two key benefits. First, even when trained purely on teleoperation data, it significantly boosts performance. Second, it enables out-of-distribution generalization through human demonstrations - allowing us to learn precise embodiment-specific skills from robot data and then *generalize* these learned skills across novel objects and compositions of these skills.**

I. Introduction

The dominant paradigm today for teaching robot policies to manipulate their environment is imitation learning (IL). Given a sufficiently large number of demonstration trajectories, imitation learning algorithms can learn precise, complex behaviors [5, 34]. However, to generalize to novel objects and environments, they require massive amounts of data, typically collected through *teleoperation*, where a human operator controls a robot to complete a given task [3, 11, 1]. This recipe, while simple, is expensive and labor-intensive to scale.

Human demonstrations are a promising way to collect large amounts of data much cheaper and faster, either through the use of web videos of humans interacting naturally with their environment [7, 8], or through task-specific demonstrations [12]. However, human data is *passive i.e.,* it does not provide any action labels with which to train imitation learning algorithms. Recent works use human demonstrations in many ways - one line of research uses vast amounts of web-scale data to learn general motion priors, and then finetune on a small amount of robot data (a few hundred trajectories) to "specialize" it to a specific embodiment [6, 2, 29]. Although these methods exhibit strong semantic generalization, they are often limited in terms of skill *dexterity*, as they cannot extract clean action labels from human web videos. Another line of work uses task-specific demonstrations collected by a human and *retargets* the human hand - extracting an action label from the human demo, which is then used to train a policy [12, 21, 9]. The most common instantiation of this paradigm involves using off-the-shelf hand pose estimators [20, 18] and mapping the detected hand keypoints to those of the robot end effector. However, using these "pseudo-actions" to train the policy depends on quality of retargeting, and poor retargeting can cause the policy to exhibit undesirable behavior - *an embodiment gap.*

We consider the setting where we have demonstrations on a set of objects and tasks (**in-distribution**) but our policy needs to be deployed on a novel task / object for which robot data is not available, but human demonstrations are (**out-of-distribution (OOD)**). Prior work looks at *enhancing* performance by supplementing small amounts of robot data with human demonstrations on the same task, while we consider out-of-distribution generalization. Assuming access to human demonstrations for such a deployment scenario, how do we adapt our policy to generalize its learned skills?

We approach this problem through the lens of a hierarchical policy framework. We decompose the policy into high-level goal prediction and low-level goal-conditioned policy [26]. We train a high-level goal prediction network that learns to predict the spatial pose of the end-effector at the end of a sub-goal, whereas the low-level policy is conditioned on the predicted goals to execute actions. We extend standard RGB-based imitation learning algorithms to condition on these 3D goals by projecting them to the image plane and representing them as *end-effector heatmaps*. **Our key insight is that this hierarchical factorization with spatial goals enables the high-level policy to learn *task-specific embodiment-agnostic* goals, whereas the low-level goal-conditioned policy learns *task-agnostic embodiment-specific* skills**. Critically, we can train the high-level policy on robot/human video as goals are embodiment-agnostic, whereas the low-level policy is trained purely on robot data to learn precise manipulation and goal-following behavior.

We find that this factorization yields two benefits: First, we show that even with only robot data, the hierarchical policy framework improves performance on in-distribution tasks. Second, through human demonstrations, we show that the framework improves OOD performance as the goal prediction transfers whereas action execution is unaffected by the human demonstrations.

Thus, we present **GHOST**, a hierarchical policy learning framework for training visuomotor policies that can generalize skills through human demonstrations. GHOST features: a) a 3D high-level goal prediction network that takes in RGB-D images to predict a distribution of sub-goal end-effector poses. b) a low-level policy conditioned on goals represented as heatmap projections of the predicted end-effector sub-goals.

Our contributions are as follows:

- A hierarchical framework for training policies on heterogeneous data sources that improves performance on in-distribution tasks and generalizes to out-of-distribution variations of the same tasks.
- A novel architecture for 3D goal prediction, aggregating RGB-D information to model distributions of sub-goals.
- A cross-embodiment spatial goal representation using *end-effector heatmaps* that enables effective goal-conditioning of image-based policies.

II. Related Work

A. Learning from Demonstrations

1) Imitation Learning: : Imitation Learning (IL) is a framework in which an agent is trained to perform sequential decision-making tasks by mimicking the behavior of an expert, from whom demonstrations are provided. While conceptually simple, it has gained widespread popularity in robotics due to the surprising complexity of behaviors that can be learned, given enough data. Recent work in this direction has been based on developing deep neural architectures that can learn dexterous and reactive policies from raw demonstration data, such as Zhao *et al.* [34], and Chi *et al.* [5] and Ze *et al.* [31]. When scaled up through large dataset collection endeavors [11, 17], much larger models can be trained and exhibit strong generalization behavior [4, 3]. However, these methods are fundamentally bottlenecked by the requirement of a large demonstration dataset, which can be unsafe, labor-intensive and expensive to collect at scale.

2) Hierarchical Imitation Learning: : Hierarchical Imitation Learning factorizes the IL framework into a high-level "planner" and a low-level "control" policy. Previous work has primarily focused on learning high- and low-level policies from robot data [15, 28, 14]. More recently, Wang *et al.* [26]

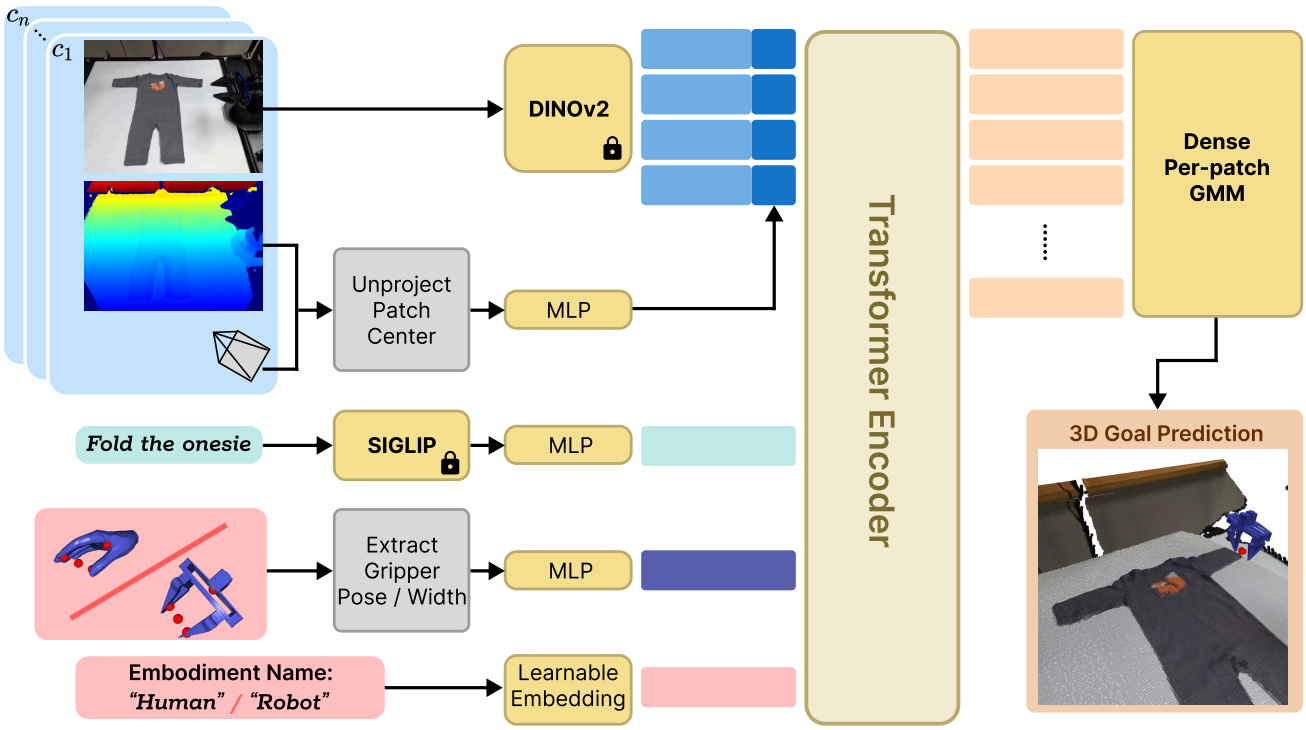

Fig. 2: **GHOST** High-level goal prediction architecture: RGB-D observations from multiple cameras are processed with a DINOv2 encoder, with patch tokens augmented by 3D coordinates. Additional context (gripper state, language embedding, embodiment name) is encoded via separate MLPs. An encoder-only transformer processes all tokens, with each patch predicting GMM parameters of a 3D goal distribution over end-effector keypoints.

demonstrated the effectiveness of a hierarchical policy to generalize to numerous articulated objects by representing sub-goals as end-effector poses. However, they do so only in the context of a single task (articulated object opening) and rely only on geometry as input to maximize sim2real transfer. We demonstrate a more general framework that can be applied to arbitrary tasks. MimicPlay [25] also use a hierarchical imitation learning framework where the high-level policy is learned from human play data to learn a latent plan over 3D end-effector trajectories. In contrast, we represent our goals as discrete end-effector goal configurations and use task-specific human demonstrations instead of play data.

B. Learning from Human Demonstrations

1) Learning policies: : With the rapid improvement in the accuracy of hand pose estimators [18, 20, 30], one direction of work focuses on tracking hand motions in human video demonstrations and explicitly representing them as robot actions in end-effector space, *i.e., retargeting* the human hand to the end-effector pose of the robot. Kareer *et al.* [10] train a unified human-robot policy by modifying ACT [34] to predict both joint positions (for robot data) and end-effector poses (for human and robot data). Ren *et al.* [21] and Haldar and Pinto [9] learn keypoint prediction models from human and/or robot demonstrations, and at inference, these are retargeted to an end-effector action based on rigid body constraints. While Haldar and Pinto [9] overcome the visual human-robot domain gap by operating purely in point-space, Lepert, Fang and Bohg [12] overlay a virtual robot onto the human arm to minimize the visual gap when training a policy from retargeted human data. As these methods aim to directly learn a policy from human demonstrations, they use minimal (or zero) robot data, but rely on the quality of hand pose estimation, and that the collected human data consists of trajectories that are executable by the robot.

2) Learning goals: : On the other hand, another line of work aims to learn "goals" and "plans" from human video, which are then used to condition a policy (trained on robot data) to guide its behavior. Bharadhwaj *et al.* [2] and Wen *et al.* [27] learn explicit track prediction models that learn to predict future trajectories of pixels conditioned on a goal image/language instruction. These track predictions are then used to condition the robot control policies to execute a given task. Collins *et al.* [6] and Ye *et al.* [29] learn *latent* dynamics from large amounts of web-scale video, with the former learning latent tracks and the latter learning latent actions through a VQ-VAE [24]. These latent dynamics are then mapped to robot actions through a separate inverse dynamics model, trained with a small amount of robot demonstrations. Our work is closest to MimicPlay [25], which learns a latent distribution over hand trajectories collected through play data, with a key distinction being that they rely on a video prompt of the task being provided at test-time, whereas we only require a language goal to be provided.

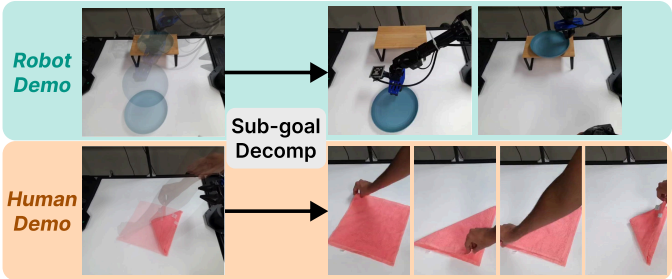

Fig. 3: Sub-goal decomposition for robot and human demonstrations. **Top**: Robot teleop demonstrations with sub-goals automatically extracted at gripper state transitions. **Bottom**: Human demonstrations with manually annotated sub-goals. Each sub-goal represents a semantically meaningful transition in the skill execution.

III. Problem Statement and Assumptions

Task Definition. We consider manipulation tasks that can be decomposed into sequential phases (*sub-goals*). We denote a *skill* as a manipulation capability described through a language instruction (*e.g.,* grasping, placing, folding) that can be instantiated across different objects and contexts [4]. Each task consists of one or more sub-goals representing transitions in task execution through a skill (*e.g.,* a pick and place task would be decomposed as pick $\rightarrow$ place).

Data. We assume access to two datasets:

- $\mathcal{D}_{robot} = \{(o_t, \text{aux}_t, a_t, e_t, g)\}$: Robot teleoperation demonstrations with workspace cameras o_t, auxiliary robot observations aux_t (*e.g.,* wrist cameras, proprioception), actions a_t, end-effector poses e_t and a language goal g, where t denotes the timestep of the observation. We also identify the timesteps where a sub-goal in a demonstration is achieved $s \in \mathcal{S}$ through heuristics (*e.g.,* gripper state changes).
- $\mathcal{D}_{human} = \{(o_t, e_t, g)\}$: Human demonstrations with camera observations o_t and hand poses e_t, derived using an off-the-shelf hand pose estimator [20, 30]. Sub-goal timesteps are manually annotated. Importantly, we note that $\mathcal{D}_{human}$ lacks action labels a_t.

We consider $\mathcal{D}_{robot}$ as **in-distribution** (ID) - demonstrations on objects/tasks for training the low-level policy. $\mathcal{D}_{human}$ contains **out-of-distribution** (OOD) demonstrations of the same skills on novel objects and object combinations, and compositions of learned skills in novel contexts.

Assumptions.

- ≥ 1 calibrated RGB-D cameras viewing the workspace for 3D goal prediction.
- The human hand remains visible during the demonstration.
- Human demonstrations use similar grasp types as the robot gripper to minimize the embodiment gap.

Problem Formulation. We learn a hierarchical visuomotor policy $\pi = [\pi_{hi}, \pi_{lo}]$:

- High-level policy $\pi_{hi} : \{o_t, e_t\} \rightarrow e_s$ predicts the end-effector pose e_s at the next sub-goal timestep s, trained on $\mathcal{D}_{robot} \cup \mathcal{D}_{human}$.
- Low-level policy $\pi_{lo} : (o_t, \text{aux}_t, e_s) \rightarrow a_t$ predicts actions conditioned on the goal e_s, trained with imitation learning on $\mathcal{D}_{robot}$.

This decomposition enables π_{lo} to learn precise robot control while π_{hi} takes advantage of human demonstrations for OOD generalization. Note that π_{hi} requires only workspace cameras o_t present in both datasets, allowing π_{lo} to exploit auxiliary sensors aux_t unavailable in human data.

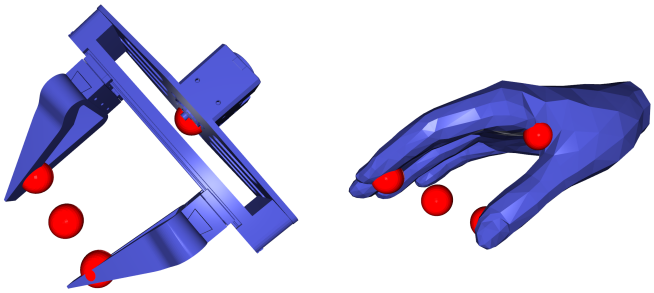

Fig. 4: End-effector representations for robot and human demonstrations. Instead of explicitly retargeting the gripper pose to a translation and SO(3) rotation, we represent gripper poses as sets of 3D keypoints (red spheres): gripper base, fingertip locations, and grasping center. **Left**: Parallel-jaw gripper. **Right**: MANO hand.

IV. GHOST

A. Data Collection and Processing

Robot Data. We collect demonstrations through teleop across N different task variants, each demonstrating the same skill (*e.g.,* pick-and-place), to provide in-distribution coverage. We extract sub-goals $s \in \mathcal{S}$ automatically by identifying the timesteps where the gripper state changes from open $\rightarrow$ close and vice versa.

Human Data. We collect human demonstrations of the *same skill* in a novel setting, using either new objects or novel applications of the skill. We track 3D hand poses H_t using off-the-shelf hand pose estimators [20, 30]. We resolve the weak-perspective scale ambiguity of the detection by segmenting the hand using Grounded-SAM [22] and scaling with the observed depth of the hand. We manually annotate sub-goals at the timesteps where a sub-goal terminates. Figure 3 shows examples of our sub-goal decomposition for both robot and human demonstrations.

Representing the end-effector pose: Following prior work, we represent the end-effector pose as a set of K 3D points instead of a position and SO(3) rotation [26]. For robot data, we sample a point cloud P_R from the gripper mesh at each time-step and represent the end-effector pose e_t as a set of 4 3D points from P_R, located at the base of the gripper, the tips of the two gripper fingers and the grasping center. Similarly, for human data, we extract hand point clouds P_H from the

MANO [23] mesh of the hand pose H_t and select 4 3D points - corresponding to points on the palm, tips of the thumb and index finger, and the grasping center. Figure 4 illustrates the keypoint locations for both gripper types. Thus, we extract a unified 3D end-effector representation e_t for human and robot data.

B. Policy Learning

1) High-Level Goal Prediction Policy: The high-level policy π_{hi} predicts the end-effector pose $e_s = \{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3, \mathbf{p}_4\}$ at sub-goal time step s, where the 4 points represent 3D positions disambiguating the end-effector [26]. We parameterize π_{hi} as an encoder-only transformer operating on RGB-D observations from C cameras. The architecture is shown in Figure 2.

Each RGB-D image is processed independently with a frozen DINOv2 [16] encoder, producing $256 \times C$ patch tokens. Each patch token is augmented with a 3D coordinate $[x, y, z]$ obtained by transforming the patch-center pixel using the camera parameters and depth. Additional tokens encode the current gripper pose (computed from e_t by extracting the $[x, y, z]$ of the grasping center, the 6D rotation [35] and the width of the gripper) and the task language embedding (with SigLIP [32]). Finally, we also include a learnable token specifying the embodiment that performs the demonstration. All tokens are projected to the transformer's feature dimension through separate MLPs.

To handle the multimodality inherent in the demonstration data [25], we model the goal distributions as a dense per-patch Gaussian Mixture Model (GMM). Each patch token predicts: (1) a mixing weight w_i, and (2) four 3D residual vectors $\{\Delta p_1, \Delta p_2, \Delta p_3, \Delta p_4\}$ relative to the patch center. The global goal distribution is a mixture of $256 \times C$ Gaussians (one per patch) with fixed variance σ^2.

Training. We train π_{hi} by minimizing the negative log-likelihood on sub-goal transitions from $\mathcal{D}' = \mathcal{D}_{robot} \cup \mathcal{D}_{human}$:

$$\mathcal{L}_{hi} = \mathbb{E}_{(o_t, e_s) \sim \mathcal{D}'} \left[-\log \left(\sum_{i=1}^{256 \times C} w_i \mathcal{N}(e_s \mid \mu_i, \sigma^2) \right) \right]$$

where μ_i is the predicted goal (patch center + residuals) for patch i.

Inference. We sample a patch i from the categorical distribution on mixing weights $\{w_i\}$, then use the corresponding mean prediction μ_i as the residual vector to be added to the patch center to obtain the predicted goal. Figure 5 visualizes the top-weighted GMM components in two camera views, showing the spatial distribution over the predicted goals.

2) Low-Level Goal Conditioned Policy: We instantiate π_{lo} as a Diffusion Policy [5], which generates chunks of action $\{a_t, ..., a_{t+H}\}$ by denoising conditioned on observations.

Goal Conditioning. The 3D goal e_s is projected onto the image plane of each camera using the camera parameters, yielding sparse 2D coordinates $\mathbf{p}_i \in \mathbb{R}^2$ for $i \in \{1, 2, 3\}$. These are converted to dense *end-effector heatmaps* H where each channel c encodes the pixel distance field from keypoint $\mathbf{p}_c$:

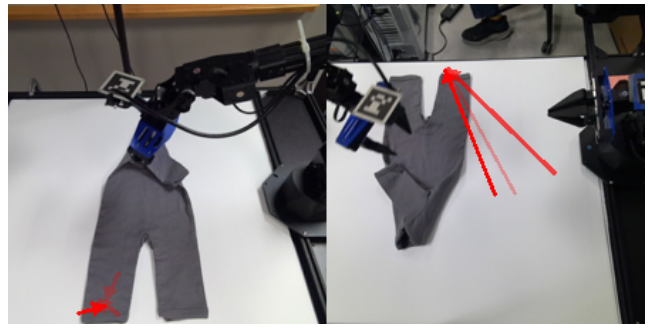

Fig. 5: High-level policy GMM predictions at inference. We visualize the top-3 mixture components per camera view, with the opacity encoding mixing weight w_i and arrows showing projections of 3D residuals Δp from patch centers.

$$H_{xy}^{(c)} = \sqrt{\frac{\|\mathbf{x}_{xy} - \mathbf{p}_c\|_2}{d_{\max}}}$$

where $\mathbf{x}_{xy}$ denotes the pixel coordinates, $d_{\max} = \sqrt{h^2 + w^2}$ normalizes by the image diagonal, and we use the square root for steeper gradients near the targets. The resulting 3-channel heatmap per camera is provided as additional input to the policy via separate image encoders, preserving spatial structure while densifying the sparse goal representation. This process is illustrated in Figure 6.

Training. We use the standard diffusion loss, training on $\mathcal{D}_{robot}$:

$$\mathcal{L}_{lo} = \mathbb{E}_{\mathcal{D}_{robot}, \epsilon} \left[\|\epsilon - \epsilon_\theta(a_t^{\text{noisy}}, o_t, \texttt{aux}_t, \text{heatmap}(e_s))\|^2 \right]$$

We train π_{hi} and π_{lo} independently, using ground-truth goals from demonstrations e_s when training π_{lo}.

Inference. At deployment, we execute π_{hi} and π_{lo} sequentially, where π_{hi} predicts a subgoal conditioned on the current observation and π_{lo} predicts one action chunk conditioned on the goal prediction.

Fig. 6: End-effector heatmap generation for goal conditioning. Predicted 3D keypoints are projected to 2D coordinates, and then converted to dense distance field heatmaps that encode spatial proximity to each keypoint.

V. Experiments and Results

We evaluate whether human video enables skill generalization in hierarchical imitation learning. Our experiments

Task	Data Source	# Demos	Generalization Type
Pick-and-Place			
`plate-on-table`	Robot	20	—
`plate-in-bin`	Robot	20	—
`mug-in-bin`	Robot	20	—
`mug-on-table`	Human	20	Object combination
`octopus-on-table`	Human	20	Object category
Cloth Folding			
`fold-onesie`	Robot	33	—
`fold-shirt`	Robot	50	—
`fold-onesie-ood`	Human	17	Object instance
`fold-onesie-reverse`	Human	25	Object instance + Skill composition
`fold-towel`	Human	50	Object category + Skill composition
LIBERO-Object			
`LIBERO-Object-9`	Robot	450	—
`LIBERO-Object-heldOut`	Robot	50	Object category

TABLE I: Task overview. Blue: ID tasks. Orange: OOD tasks requiring generalization.

Method	Success Rate (%)		
	`plate-on-table`	`mug-on-table`	`octopus-on-table`
DP	**60**	**5**	0
Hierarchical DP w/out Human	**70**	0	0
GHOST (Ours)	**60**	**32.5**	**20**

TABLE II: Pick-and-place results. Best: **shaded**, second-best: **bold**. Blue: ID. Orange: OOD.

Task	Method	Success Rate (%)				
		1-step	2-step	3-step	4-step	Final
`fold-onesie`	DP	**100**	70	30	0	0
	Hierarchical DP w/out Human	**100**	**100**	**100**	**100**	**80**
	GHOST (Ours)	**100**	**100**	**90**	**90**	**70**
`fold-onesie-ood` (Novel Object Instance)	DP	40	40	10	0	0
	Hierarchical DP w/out Human	**100**	**100**	60	50	40
	GHOST (Ours)	**100**	**100**	**90**	**70**	**50**
`fold-onesie-reverse` (Novel Object Instance + Skill Composition)	GHOST (Ours)	100	100	70	20	0
`fold-towel` (Novel Object Category + Skill Composition)	GHOST (Ours)	60	—	—	—	5

TABLE III: Cloth folding results. k-step = first k folds completed. Final = all folds completed (5-step for the **`fold-onesie`** tasks, 2-step for **`fold-towel`**). Best: **shaded**, second-best: **bold**. Blue: ID. Orange: OOD.

Method	Success Rate (%)	
	`LIBERO-Object-9`	`LIBERO-Object-heldOut`
DP	**88**	0
Hierarchical DP	**90**	**24**

TABLE IV: LIBERO-Object results. Best: **shaded**, second-best: **bold**. Blue: ID. Orange: OOD.

test three hypotheses: (1) hierarchical policies improve in-distribution performance even without human data, (2) human demonstrations enable transferring learned skills to novel object instances and categories, and (3) human demonstrations enable *compositional generalization* and enable applying learned skills in novel sequences and contexts.

A. Tasks and Evaluation Protocol

Table I summarizes our evaluation tasks. We collect robot demonstrations on in-distribution (ID) tasks to train both π_{hi} and π_{lo}, and human demonstrations on out-of-distribution (OOD) variations to train only π_{hi}. For LIBERO-Object [13], we use robot demonstrations for both the ID and OOD tasks.

- **Pick-and-Place:** Grasp a specified object (plate, mug, octopus) and place on a target (table, bin). Robot demos cover plate and mug on the two targets; human demos provide OOD object-target combinations. The task is considered a success if the object is placed stably on the target, objects placed successfully but knocked off during reset are given a score of 0.5.
- **Cloth Folding:** Long-horizon deformable manipulation

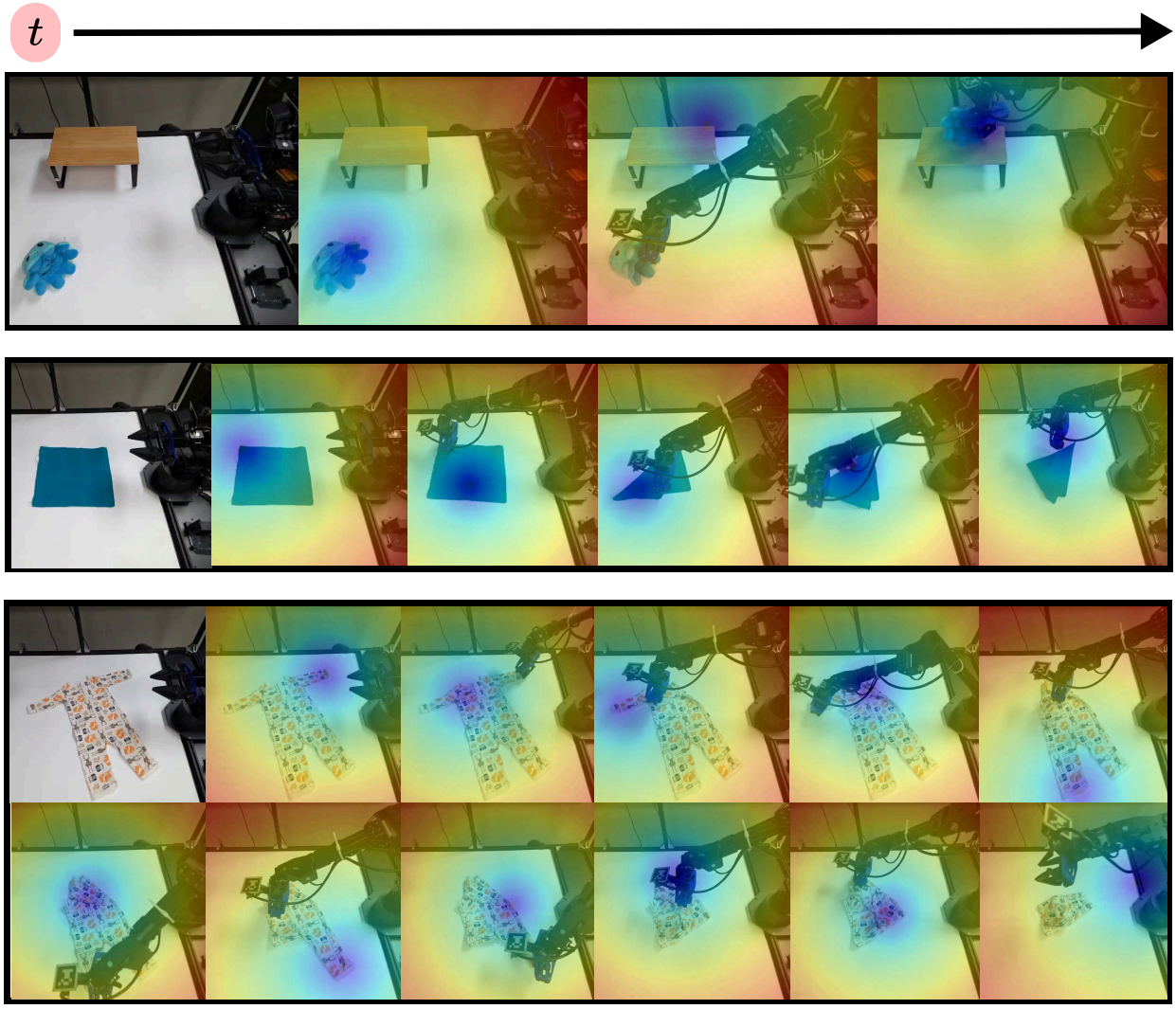

Fig. 7: Qualitative results on OOD tasks. **Top**: `octopus-on-table` (novel object category). **Middle**: `fold-towel` (novel category + skill composition). **Bottom**: `fold-onesie-ood` (novel instance). We visualize rollouts by overlaying a colormap on the projection of π_{hi}'s goal predictions at each timestep.

requiring multiple sequential folds. Robot demos on folding onesies and shirts; human demos on novel instances (`fold-onesie-ood`), composing the skill in novel sequences (`fold-onesie-reverse`), and applying the skill on entirely novel categories (`fold-towel`). k-step success indicates first k folds completed successfully

- **LIBERO-Object:** Standardized pick-and-place benchmark [13] with 10 objects. We train π_{lo} on 9 objects, π_{hi} on all 10, and evaluate transfer to the held-out object. Success is reported by the simulation environment.

B. Baselines

We evaluate GHOST against two baselines: (1) Diffusion Policy (DP) [5], a flat policy trained on robot data, and (2) Hierarchical DP, our hierarchical framework trained only on robot data without human demonstrations. All policies are trained as multi-task policies with language conditioning. For our hierarchical framework, language conditioning is only provided for the high-level policy. For the Diffusion Policy baseline, we add language conditioning by appending the text embedding to the global conditioning vector.

C. Implementation

We train all DP variants for 300k steps with color jitter and random crop augmentations. The high-level policy is trained for 100 epochs with additional augmentations: adding noise to the gripper pose, blur, grayscale, and token dropout augmentations. At inference, we synchronously execute π_{hi} and π_{lo} - the low-level predicts action chunks that fully execute before the next high-level inference step.

D. Results

Do hierarchical policies improve performance on in-distribution tasks? Yes. Hierarchical DP w/out Human outperforms flat DP on ID tasks (Table II, Table III, Table IV). We see modest performance gains on simpler tasks

(70% vs 60% on `plate-on-table` and 90% vs 88% on `LIBERO-Object-9`). However, we see a significant increase on `fold-onesie` (80% vs 0% final success), indicating that the hierarchical factorization is especially critical for long-horizon tasks. GHOST achieves comparable ID performance (60% on pick-and-place, and 70% on `fold-onesie`), suggesting that training π_{hi} with human demonstrations does not meaningfully degrade in-distribution behavior.

Do human demonstrations enable transferring learned skills to novel object instances and categories? Yes. Human demonstrations unlock meaningful OOD transfer of learned skills to novel objects. GHOST achieves 32.5% success on `mug-on-table` and 20% on `octopus-on-table` vs 0% for both baselines (Table II). On `fold-onesie-ood` (novel instance), GHOST achieves 50% final success vs 40% for Hierarchical DP w/out Human and 0% for flat DP (Table III). The hierarchical baseline's complete failure on pick-and-place OOD (despite 70% ID performance) confirms that human demos are critical for generalizing goal prediction. For LIBERO, we do not use human demonstrations but demonstrate that by training π_{hi} on `LIBERO-Object-heldOut`, hierarchical modeling enables 24% OOD success vs 0% for flat DP (Table IV).

Do human demonstrations enable *compositional generalization* of learned skills to novel object instances and categories? Partially, but performance degrades sharply with horizon (Table III). On `fold-onesie-reverse` (novel instance + skill composition), GHOST maintains 100% success through 2 steps and 70% at 3 steps but drops to 0% for final success. Similarly, on `fold-towel` (novel category + skill composition), the degradation is more severe: 60% 1-step success drops to 5% final. The visual domain gap between human and robot data appears to compound across sub-goals, suggesting that the high-level policy struggles to maintain accurate goal predictions as errors accumulate.

VI. Limitations and Future Work

Although GHOST shows promising results in generalizing learned skills across multiple axes in a sample-efficient setting, there are numerous limitations. Like other methods that use hand pose estimators on human demonstrations [12, 21], our approach is limited by the quality of the estimated hand pose, and suffers in situations where hand pose estimation is weak, such as occlusions.

We extract sub-goals through heuristics for robot data (gripper open/close), which may not be optimal or even valid for tasks requiring continuous manipulation skills (pouring, stirring), and through annotation on human demonstrations, which imposes an additional cost on the collection of demonstrations. Related work on automatic sub-goal discovery [19, 33] is an interesting direction for future research on mining sub-goals from collected human and robot demonstrations.

While GHOST is able to generalize to novel object instances and compose learned skills in novel sequences, we observe a steep drop in performance as the horizon of the task lengthens, which we observe to be caused by the visual domain gap for the high-level policy. Existing work using human and robot data seeks to learn embodiment-invariant visual representations [25, 21] to minimize this gap and represents an important research direction for future work.

Lastly, the promise of this hierarchical policy factorization lies in its ability to scale high-level and low-level policies independently on large human and robot datasets. This allows learning both generalizable goal prediction and goal-following behavior, and is an exciting path forward towards learning generalist robot policies.

VII. Conclusion

We present GHOST, a hierarchical imitation learning framework that decouples embodiment-agnostic goal prediction from embodiment-specific action execution. This factorization enables two key benefits: improved in-distribution performance through explicit sub-goal modeling, and out-of-distribution generalization via human demonstrations without requiring action labels from human data.

We introduce a 3D goal prediction architecture that processes multi-view RGB-D observations and predicting dense per-patch Gaussian mixture models over end-effector poses. We represent goals as unified 3D keypoint sets across embodiments and condition the low-level policy through end-effector heatmaps. This representation bridges the gap between human and robot demonstrations without requiring explicit action retargeting.

By training π_{hi} on heterogeneous human and robot data and π_{lo} purely on robot demonstrations, we generalize learned skills across novel object-context combinations (32.5% success versus 0% for baselines on out-of-distribution pick-and-place) and meaningful progress on compositional skill generalization (60% task progress on novel cloth folding instances).

Acknowledgments

We would like to thank Alexis Hao, Mino Nakura, Kallol Saha and Pratik Bhowal for helpful discussions and assistance with data collection. We also thank the other members of the R-PAD lab for their feedback.

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